USED MEDIAS LLC

EMBEDDED SYSTEMS DIVISION

Engineering Documentation

cyd-os

DOCUMENT	REVISION	STATUS	CLASSIFICATION
cyd-os	—	—	Internal

Purpose of this set

These reports record the design of cyd-os in enough detail that the project can be picked up cold — by someone else, or by the author after six months away — without re-deriving decisions from the source.

Each report states **what was decided, what it was measured against, and what remains unverified**. Claims that were tested on hardware are marked as such. Claims taken from documentation or reasoning are marked separately, because on a from-scratch kernel the difference between "this is true" and "this should be true" is the difference between a working boot and a silent reboot.

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ID	Title	Covers
UM-CYDOS-013	Milestone 5 Verification Report	Application table and lifecycle, three-level scheduling, the shell, and the escape attempt
UM-CYDOS-014	Locking Primitives	Critical sections vs blocking mutex, task blocking and idle, and a measured starvation defect
UM-CYDOS-015	Display Driver	ILI9341 over bit-banged SPI, span rendering with no framebuffer, and measured throughput
UM-CYDOS-016	Display Syscalls, and a Total System Freeze	Per-application viewports, pointer discipline, measured containment, and a yield that stopped the clock
UM-CYDOS-017	Touchscreen, and a Verification Method That Failed Three Times	XPT2046 over PENIRQ, the GPIO two-bank bug, calibration by controlled input, a capture that erased its own evidence, and input confinement

Reading order

New to the project: **001** → **002** → **004** → **003**. That gives the shape of the system, how it starts, where things live, and why the calling convention is unusual.

Picking up implementation work: **006** (what is known-good) then **007** (what is next and what it will break).

Reproducing a build: **005** alone is sufficient.

Status summary as of this revision

Item	State
Milestone 0 — kernel boots, self-checks pass	**Complete, verified on hardware**
Milestone 1 — timer interrupt, tick counter	**Complete, verified on hardware**
Milestone 2 — native task switching	**Complete, verified on hardware** — 3,400+ switches, zero corruption
Milestone 3 — heap and arena model	**Complete, verified on hardware** — all three exit criteria
Milestone 4 — bytecode interpreter	**Complete, verified on hardware** — program runs, six fault classes contained
Isolation	**Live** — every VM memory access bounds-checked, UM-CYDOS-012 §6.2
Full stack	**Running** — bytecode hosted in a preemptible native task, 3.3M instructions across 450 preemptions with zero accounting drift, UM-CYDOS-012 §6.6
Milestone 5 — multiple applications	**Complete, verified on hardware** — all three exit criteria
Roadmap M0–M5	**Complete.** Six native tasks, three scheduling levels, a shell, and an application deliberately written to escape its arena that could not
Locking	**Complete** — critical sections and a blocking mutex; heap and console both arbitrated, UM-CYDOS-014
Task blocking	**Live** — `TASK_BLOCKED` plus an idle task using `WAITI`, UM-CYDOS-014 §3
Display	**Working on hardware** — ILI9341, no framebuffer, 387 ms full-screen fill, colour order confirmed, UM-CYDOS-015
Application graphics	**Live** — `FILL` / `TEXT` / `DIMS` confined to per-application viewports; 0 escapes across 136 audited fills, UM-CYDOS-016 §5
Touch	**Working on hardware** — XPT2046 via PENIRQ, both axes calibrated by measurement, UM-CYDOS-017
Application input	**Live** — `SYS TOUCH` confined to the asking application's viewport; 81 delivered, 109,211 withheld, 0 confinement failures, UM-CYDOS-017 §8
DRAM budget	**Measured** — 158,000 B allocatable today (167,680 before `TASK_MAX` rose to 8); a full framebuffer is unnecessary, UM-CYDOS-010 §7.2
Flash cache	**Enabled** — `.rodata` mapped from flash, UM-CYDOS-011
Version control	**Initialised 2026-08-14**
JTAG debug probe	Ordered, not in hand

Item	State
Bootloader IRAM overlap	**Closed** — investigated and disproved, UM-CYDOS-004 §5
Panic handler	**Closed** — exercised deliberately with an `ill` instruction; prints EXCCAUSE/EPC and halts
Watchdogs	**Closed** — measured armed, now disabled and read back, UM-CYDOS-009 §8

> **Standing rule for interrupt handlers — clear `PS.EXCM` before calling C.** Hardware sets `EXCM` on interrupt entry, and while it is set the Xtensa zero-overhead loop-back is disabled: a hardware loop body executes once and falls through to whatever sits at `LEND`. GCC emits these loops for ordinary counted C loops, so this silently degrades **every C function reachable from a handler** — drivers and the VM interpreter included, not just the scheduler where it was found. `_handler_level3` now clears it; any future handler must do the same. UM-CYDOS-009 §6.
>
> **Standing rule for the scheduler — a yield must never defer the clock it depends on.** `task_yield()` originally wrote `CCOMPARE1 = ccount + 64` unconditionally, so any loop yielding faster than 64 cycles pushed the deadline ahead of `CCOUNT` forever, the timer interrupt stopped firing, and the whole kernel froze — the tick is what drives every context switch. Any routine adjusting a scheduler deadline must only ever move it **earlier**. UM-CYDOS-016 §3.
>
> **Standing rule for verification — when an instrument reports its own reading invalid, that outranks every value printed beside it.** Three times across two sessions a frozen marker, a sample counter stuck at an identical value, and a boot banner in a serial capture each said "this measurement is not what you think", and the plausible-looking numbers next to them were believed anyway. Two conclusions were published and later retracted as a result. Latch the quantity so timing cannot lie about it, then feed the system a controlled input rather than interpreting an uncontrolled one. UM-CYDOS-017 §6.